Iris Phenolic Liners
Continuous Heating to Model Exhaustion

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1 How long does the liner last in the calculation?

For a continuously exposed liner, the inherited independent model reaches the casing after roughly two minutes for the sustainer and two and a half minutes for the booster. Each event occurs after the casing reaches the limit of its property evidence. Actual firing heat flux, local exposure duration and measured liner loss are unknown. The times below describe a hypothetical intact backing under sustained heating. Physical motor burnthrough remains open.

	Time to final liner-row loss [s]			Property flag [s]
Case	160, full step	160, half step	320, half step	320, half step
Sustainer Al	114.95	114.21	115.10	49.49
Sustainer CF	112.18	111.42	112.30	18.34
Booster Al	151.70	151.03	151.95	87.27
Booster CF	147.05	146.36	147.27	32.71

Table 1: Independent radial calculations. The initial thicknesses are 4.7625 mm for the sustainer and 6.3500 mm for the booster. A flag marks the first 150 °C CF or 300 °C Al interface crossing. The table separates this evidence limit from complete liner-row loss.

The gas stays at 2504 K (2230.85 °C), with nominal pressure 800 psi and uncorrected film coefficient 1000 W/(m^2 K). Outgassing modifies convection. Effective hot-side emissivity is zero. The outside sees 22 °C, a film coefficient of 8 W/(m^2 K) and emissivity 0.25. The hot surface temperature evolves as the wall absorbs heat. The local liner face is exposed from time zero until exhaustion. Sustained heating is a chosen scenario.

The three columns use 160/40, 160/40 and 320/80 radial cells in the liner and casing. The full step ceiling is 0.0015625 s and the half step is 0.00078125 s. In all twelve records the final row disappears within that run's time increment. Resolution changes move the event by more than an individual time increment, so digits beyond those shown carry little meaning.

2 Liner remaining and casing temperature

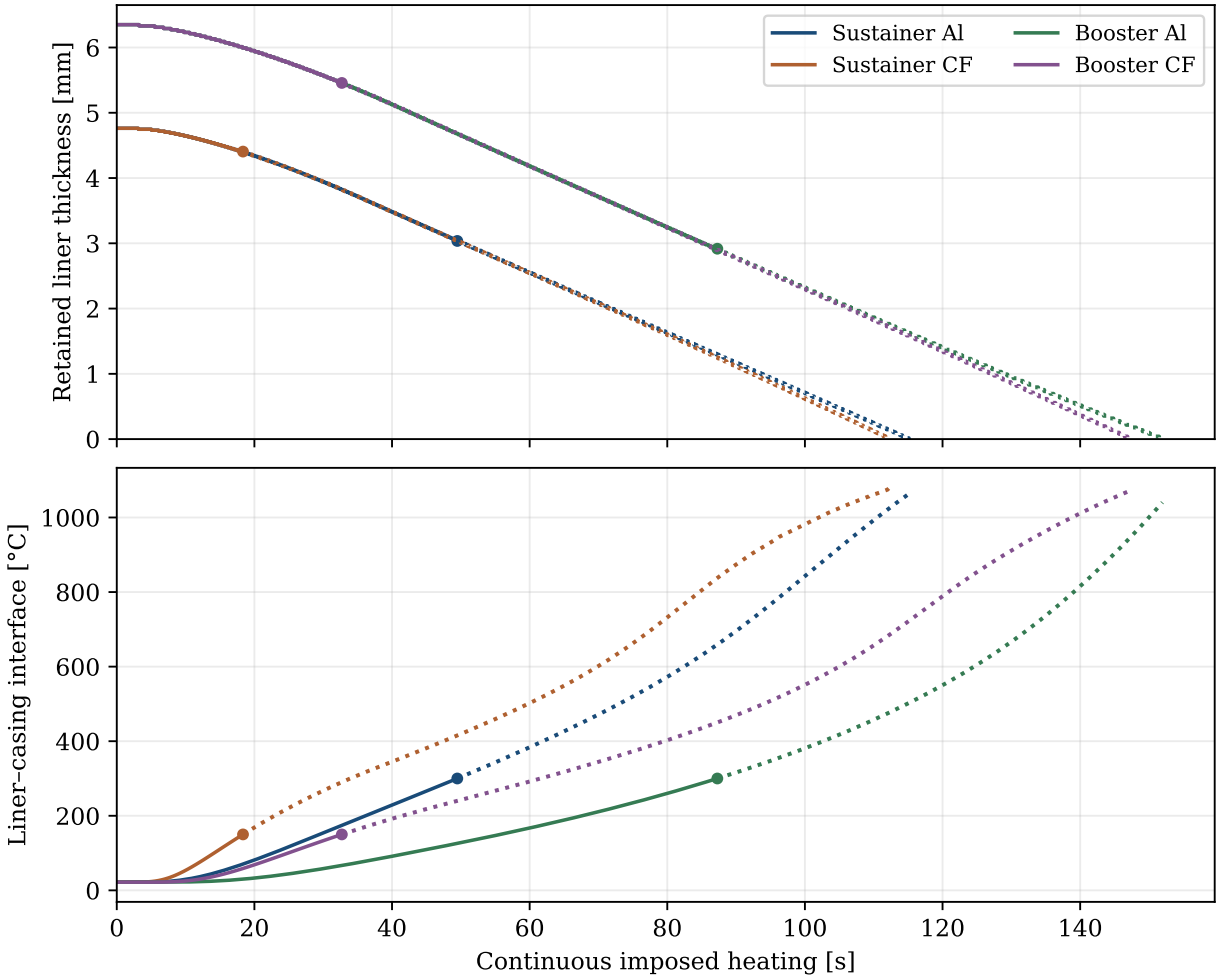


Figure 1: Independent 320/80 finite-row results. Dots mark the first casing-property flag. The dotted continuation assumes an intact casing with the inherited surrogate properties. Thickness includes retained char and is distinct from the thickness of virgin material. Similar curves overlap near ignition.

Al uses the inherited constant properties. The CF backing uses its radial conductivity and a heat-capacity correlation clamped to the inherited interval. The plotted temperatures can exceed the materials' supported range by hundreds of degrees. Integrity of a hot pressurized case, resin decomposition, bond failure and local cracks require additional evidence. The previous 7 s pulse and cooling results, native ANSYS verification and complete MAPDL and Fortran listings are in the companion native report. Earlier rejected endurance jobs are excluded from every comparison here.

3 What changes with time step and mesh?

Time comparisons halve the time step at fixed 160/40 mesh. Mesh comparisons double the number of cells in both materials at the same half step. All other physics inputs match between each pair. A signed time change is the finer result minus the coarser result.

	Event change [s]		Remaining thickness [mm]		
Case	Time	Mesh	Time diff.	Mesh diff.	Target
Sustainer Al	-0.74	+0.89	0.0595	0.0595	0.0476
Sustainer CF	-0.76	+0.88	0.0595	0.0595	0.0476
Booster Al	-0.67	+0.93	0.0397	0.0595	0.0635
Booster CF	-0.69	+0.91	0.0397	0.0595	0.0635

Table 2: Thickness columns give the largest absolute difference on a shared 0.5 s grid up to the first exhaustion. Every row transition was archived, so sampling uses the retained row count at each comparison time.

The largest change in event time is 0.78 percent of the finer calculation. 4 of eight comparisons meet the intermediate-thickness target and 0 of eight pairs have overlapping final-event brackets. Both sustainer pairs miss the 0.047625 mm target. Both booster pairs meet their 0.0635 mm target on the shared grid and over the full common history. Halving the step advances the event in all four cases, while doubling the mesh count delays it. Comparing only the first and third columns would partially cancel those effects. Each row retains its full thermal capacity during partial consumption until its residual-mass criterion removes it. The candidate in the next section replaces that discrete handoff with continuous front motion.

A separate thin-coupon test paused the independent worker after the second row loss and casing flag. A fresh controller then recovered the same complete history, final-event bracket and byte-identical terminal checkpoint as uninterrupted integration. The audit reconstructs final mass from the saved states, checks each row transition, and matches source and input fingerprints across all twelve records. Maximum cumulative relative mass error is 1.54e-14. Maximum relative residual of the inherited discrete step energy equation is 9.22e-11. The new material and wall equations below use a separate open-system energy ledger. A new 585-node native conduction verification input has been sealed for ten-core ANSYS execution after the active job. The candidate still needs a coupled transient check, then independent time and 160/320/640 mesh studies and comparison against fresh native verification. Measured heating and recession, high-temperature backing data and sensitivity to surface chemistry remain necessary for the physical interpretation.

4 A moving surface with separate virgin, char and gas inventories

The candidate transports the masses of virgin phenolic M_v, char M_c, and their combined sensible enthalpy H_s in each cylindrical cell. A cell has volume $(r_o^2 - r_i^2)/2$ per radian per metre of axial length. Conservative overlap integration in $r^2/2$ returns both the retained and swept inventories as the face advances. In the equations below, M and H may refer either to a physical cell or to these quantities per radian per metre, provided every source uses the same convention.

4.1 Bulk reaction and reconstruction

We use a common reference $T_* = 22\,^\circ\mathrm{C}$ for virgin solid, char and gas. The inherited table defines piecewise-linear specific heat capacities. Let $h_i(T) = \int_{T_*}^{T} c_{p,i}(u)\,du$ for $i = v, c$ and $h_g(T) = 1600(T - T_*)$ J/kg when T is in degrees Celsius. The candidate holds the first heat capacity constant between 0 and 22 degrees Celsius and rejects states above the final 2476.85 degree table entry or below zero. Positive phase masses give a strictly increasing enthalpy relation $H_s = M_v h_v(T) + M_c h_c(T)$. Inverting the quadratic integral within its table interval recovers a single temperature after remapping. An exhausted cell carries zero mass and enthalpy and has no temperature to reconstruct.

Let $y = 600/1250 = 0.48$ denote the assumed char yield per kilogram of virgin material consumed. With $\bar{T} = (T^n + T^{n+1})/2$ and $k = 333\exp[-64081/(R(\bar{T} + 273.15))]$ in reciprocal seconds, the update uses

$$d = M_v^n(1 - e^{-k\Delta t}), \quad M_v^{n+1} = M_v^n - d, \quad M_c^{n+1} = M_c^n + yd, \quad m_g = (1 - y)d.$$

Mass and enthalpy use the same d. The scalar temperature solve enforces

$$H_s^{n+1} - H_s^n = Q_{\mathrm{cell}} - L_p d - m_g h_g(\bar{T}).$$

The chosen basis for $L_p = 418$ kJ/kg is virgin mass consumed at T_*. Gas leaves carrying its sensible enthalpy at generation. The phase inventories on the left already include the change in stored solid sensible heat.

4.2 Partially converted material at the wall

Let a and b be the virgin and char mass densities in the material adjacent to the gas. The current candidate assumes that residual virgin material swept by the front completes pyrolysis there and its char reacts at that same face. For speed $v \geq 0$ the outgoing surface pyrolysis gas flux is $j_g = (1 - y)av$ and the outgoing char-product flux is $j_c = (b + ya)v$, in kg/(m^2 s). Their sum equals the rate $(a + b)v$ at which the surface sweeps up solid. Writing $\alpha = b/(b + ya)$, we impose $j_c = \alpha j_{\mathrm{law}}(T_s, p)$ with the inherited kinetic and mass-transfer law. A virgin wall can heat and pyrolyse volumetrically before surface recession begins.

For a first-cell temperature T_1, define the sensible power delivered to the boundary by swept inventory as $q_{\mathrm{swept}} = v[ah_v(T_1) + bh_c(T_1)]$. Per unit hot-face area the wall balance is

$$q_{\mathrm{conv}} + q_{\mathrm{rad}} + q_{\mathrm{swept}} = q_{\mathrm{cond}} + j_g h_g(T_s) + j_c h_c(T_s) + avL_p + j_c L_c,$$

$$q_{\mathrm{cond}} = \frac{\displaystyle\int_{T_1}^{T_s} [(1 - \alpha)k_v(T) + \alpha k_c(T)]\,dT}{r_s \log(r_1/r_s)}.$$

Here $q_{\mathrm{cond}} > 0$ flows into the retained solid, r_s is the moving wall radius and r_1 the adjacent cell centre. The conduction expression integrates the cylindrical resistance at fixed local composition. The film coefficient uses total outward mass flux, including bulk gas arriving at the surface and both j_g and j_c. A bracketed scalar solve closes film heating, optional radiation, conduction and recession at the same T_s. Bulk gas has already exported its sensible heat in the bulk equation. The chosen proxy $h_c(T_s)$ for the char product and excess cost $L_c = 35$ MJ/kg together specify a provisional removal energy. Data for reaction products, recession and pyrolysis are needed to determine whether these assumptions hold for the Iris liner.

4.3 Verification to date

Tests compare the enthalpy primitive against independent quadrature and the isothermal reaction against the analytic solution. In a four-second manufactured heated cell, doubling the time steps from 8 to 16, 32 and 64 reduces temperature error against an independent ODE integration from 0.151 to 0.0376, 0.00940 and 0.00235 K. A nonuniform reacting shell with prescribed recession closes the integrated mass and energy ledgers through final-cell removal to relative errors below 2×10^{-15}. Separate manufactured tests recover imposed face temperatures for a virgin conducting wall, pure reacting char, and a mixed wall with blowing and radiation. The face equation residual is below 3×10^{-9} W/m^2 for those fixtures. These checks exercise individual material and boundary balances. The coupled transient wall, native implementation and event-time comparison remain open.

4.4 Radial heat flow between cells

On a stationary mesh the conduction operator transfers heat between adjacent cell centres through the two cylindrical half-cell resistances. For centre radii r_i, r_{i+1} and shared face r_f, the heat flow per radian per axial metre toward increasing radius is

$$F_{i+1/2} = \frac{T_i - T_{i+1}}{\log(r_f/r_i)/k_i + \log(r_{i+1}/r_f)/k_{i+1}}.$$

Backward Euler balances each cell's change in sensible heat against the difference of its two face flows and any supplied power. A two-layer steady wall reproduces the logarithmic temperature solution to within 6×10^{-13} K. A transient annular conduction test compares against an analytic Bessel eigenmode over one decay time. With the time error removed to first order, the maximum errors on 16, 32 and 64 cells are 0.118, 0.0295 and 0.00738 K. These tests exercise diffusion and the material interface at fixed geometry. Verifying the jointly evolving temperature, gas flow and surface position is the next step.

Reference for the method. Chen and Milos describe a moving-grid treatment of decomposition, energy and surface balances for a charring ablator. Their PICA data apply to a different material and environment. NASA NTRS 20130014432.

A Conservative geometry and remapping code

Below is the complete independent geometry kernel. The accompanying model directory contains the bulk reaction, boundary and conduction implementations and their verification programs.

```
"""Conservative radial geometry kernel for a future moving-surface model.

This is NOT a heat-transfer/reaction solver or a recession law. It transports
extensive states under a prescribed boundary movement. Units are per radian
and per metre of axial length; multiply by sector angle and length if needed.
"""
from dataclasses import dataclass
import numpy as np


@dataclass
class Remap:
    density: np.ndarray
    cell_integrals: np.ndarray
    removed_integrals: np.ndarray
    original_integrals: np.ndarray
    balance_residual: np.ndarray


def advance_front(radius, interface, speed, dt):
    """Locate geometric exhaustion for a prescribed constant normal speed.

    The caller must obtain speed from a verified surface mass/energy balance.
    A temperature, conversion threshold or arbitrary cutoff is not used here.
    """
    if not np.all(np.isfinite([radius, interface, speed, dt])):
        raise ValueError('Nonfinite front inputs')
    if radius <= 0 or interface < radius or speed < 0 or dt <= 0:
        raise ValueError('Invalid front inputs')
    if radius == interface:
        return interface, 0., True
    if speed == 0:
        return radius, dt, False
    to_event = (interface - radius) / speed
    if to_event <= dt:
        return interface, to_event, True
    return radius + speed * dt, dt, False


def remap(old_edges, density, new_edges):
    """Integrate piecewise-constant conserved densities over new radial cells.

    The last dimension of density is the old cell index. Other dimensions can
    hold virgin-equivalent mass, char mass, solid enthalpy and related conserved
    quantities. Remap those quantities, not temperature or alpha independently.
    Thermodynamic reconstruction and reaction/removal source terms are the
    caller's responsibility. Interface positions must be retained by the caller
    so heterogeneous material labels are not averaged into one material.

    New cells cover the retained domain [new inner radius, old outer radius].
    A one-entry new_edges array denotes exact exhaustion of this domain. The
    swept-out mass and enthalpy are returned explicitly, not silently deleted.
    """
    old = np.asarray(old_edges, dtype=float)
    new = np.asarray(new_edges, dtype=float)
    field = np.asarray(density, dtype=float)
    if old.ndim != 1 or new.ndim != 1 or len(old) < 2 or len(new) < 1:
        raise ValueError('Expected radial edge arrays')
    if field.ndim < 1 or field.shape[-1] != len(old) - 1:
        raise ValueError('Density field does not match old cells')
    if not all(np.all(np.isfinite(x)) for x in [old, new, field]):
        raise ValueError('Nonfinite remap inputs')
    if old[0] <= 0 or np.any(np.diff(old) <= 0) or np.any(np.diff(new) <= 0):
        raise ValueError('Cells must have strictly positive radial volume')
```

```
    if new[0] < old[0] or new[-1] != old[-1]:
        raise ValueError('Only␣inner␣recession␣with␣a␣fixed␣outer␣boundary␣is␣supported')
    # x = r^2/2 is the exact cylindrical volume coordinate per radian/metre.
    xold = old * old / 2
    xnew = new * new / 2
    old_integrals = field * np.diff(xold)
    cumulative = np.concatenate([np.zeros(field.shape[:-1] + (1,)),
                                 np.cumsum(old_integrals, axis=-1)], axis=-1)
    flat = cumulative.reshape((-1, len(old)))
    retained_cumulative = np.array([np.interp(xnew, xold, row) for row in flat])
    retained_cumulative = retained_cumulative.reshape(field.shape[:-1] + (len(new),))
    integrals = np.diff(retained_cumulative, axis=-1)
    removed = retained_cumulative[..., 0]
    original = cumulative[..., -1]
    residual = original - removed - integrals.sum(axis=-1)
    new_density = integrals / np.diff(xnew)
    return Remap(new_density, integrals, removed, original, residual)
```